

Exam 2026, Part 2

Exam number 45, 46 & 47

June 26, 2026

General Remarks

All members participated in the discussion of the empirical approach, interpretation of results, code review, and final editing. All parts were jointly reviewed by the group. To make the individual contributions transparent, the primary responsibility for the different parts is distributed as follows: Exam number 45: ex. 1 & 4 Exam number 46: ex. 2 & 5, Exam number 47: ex. 3.

The use of generative AI tools is documented in the submitted GAI declaration.

Exercise 1: Choice of realized variance estimator

We first load the high-frequency minute price data and filter the dataset to AAPL. The timestamp column is converted to datetime format, and we define each trading day from the timestamp.

The data contain minute-level closing prices during regular trading hours. We place AAPL prices on a fixed one-minute calendar-time grid from 09:30 to 15:59, corresponding to 390 observations per normal trading day. Missing prices are handled by previous-tick interpolation within each trading day only: if no price is observed in a given minute, the most recent price from the same day is carried forward. Before interpolation, AAPL's missing share on this grid is 0.40%. This creates a complete intraday grid, but may also introduce stale prices in minutes without new trades. We therefore do not carry prices across trading days or backfill before the first observed price of a day.

For each sampling interval $\Delta \in \{1, 2, 5, 10, 15, 30, 60\}$, prices are sampled at pre-specified timestamps relative to the market open, e.g. 09:30, 09:40 for 10-minute sampling. A bucket-based approach would instead typically use the last observed price within each interval, e.g. 09:39, 09:49 for 10-minute buckets. We use fixed-grid sampling to make the timing convention explicit and to ensure that intraday returns are aligned across stocks when constructing realized covariance matrices.

Daily realized variance is computed as:

$$RV_d^{(\Delta)} = \sum_j \left(r_{d,j}^{(\Delta)} \right)^2,$$

where $r_{d,j}^{(\Delta)}$ is the intraday log return over sampling interval.

As shown in Figure 1, average realized variance is highest at the shortest sampling intervals and decreases as the sampling interval increases. This pattern is consistent with market microstructure noise at very high frequencies, such as bid-ask bounce, price discreteness, and stale prices. At very fine sampling intervals, these frictions can inflate realized variance because observed transaction prices deviate from the latent efficient price. Sampling less frequently reduces this noise, but very long sampling intervals also discard useful intraday information.

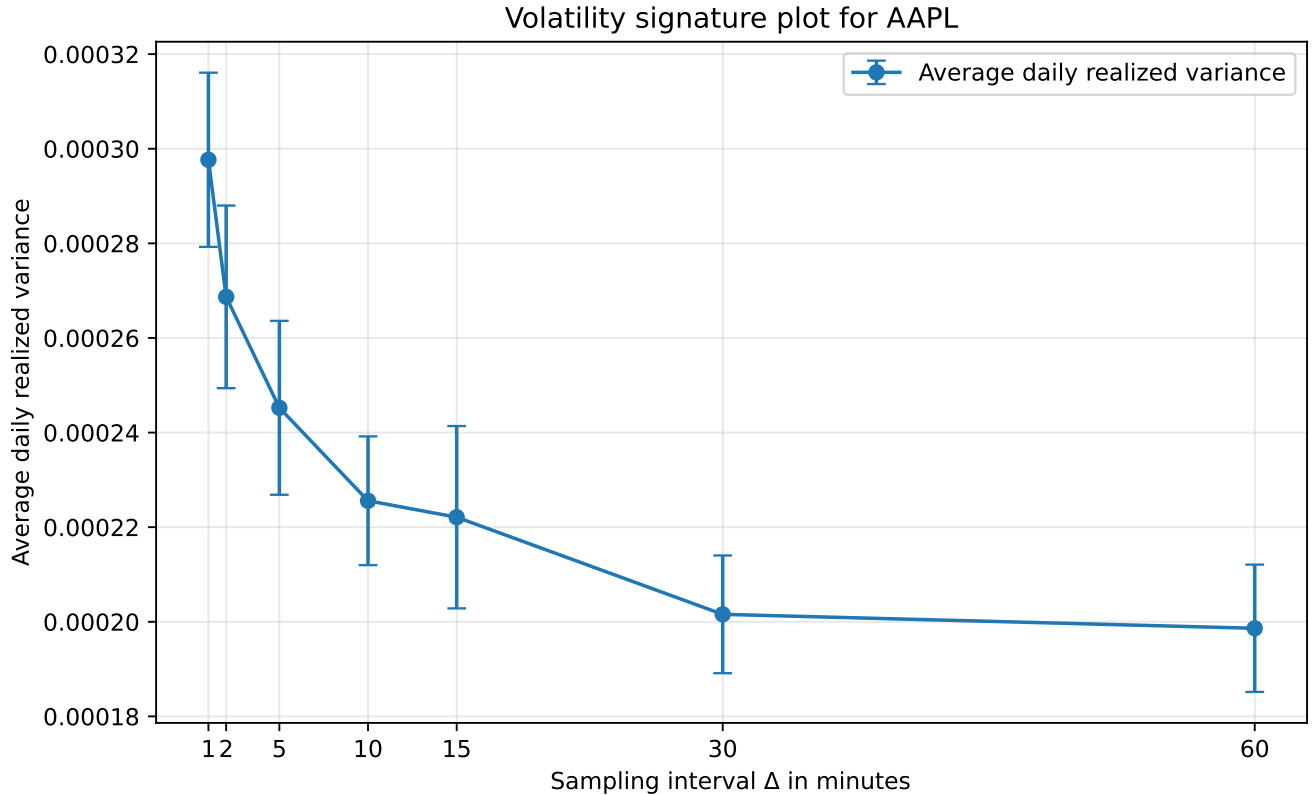


Figure 1: Volatility signature plot for AAPL with 95% confidence intervals.

The volatility signature plot shows that the average realized variance is highest at the shortest sampling intervals and falls substantially when moving from 1-minute to 5-minute sampling. This is consistent with the presence of market microstructure noise at very high frequencies. In the lecture slides, the observed log price is written as the latent efficient log price plus a noise component, $Y_{i\Delta_n} = X_{i\Delta_n} + U_{i\Delta_n}$, and the noise-driven fraction of return variance increases as Δ_n becomes small. Sampling less frequently therefore reduces the influence of bid-ask bounce, price discreteness and stale prices.

We choose the standard realized variance estimator based on 5-minute log returns for the remainder of the analysis:

$$RV_d^{(5)} = \sum_j \left(r_{d,j}^{(5)} \right)^2.$$

This choice is also consistent with the empirical realized-volatility literature, where 5-minute returns are commonly used as a practical compromise between using high-frequency information and limiting market microstructure noise. Andersen and Bollerslev discuss the use of high-frequency intraday returns to obtain improved volatility measurements and refer to empirical analysis based on five-minute returns. At the same time, 5-minute sampling leaves approximately $390/5 - 1 = 77$ intraday returns per trading day. Since the subsequent realized covariance matrix is 30-dimensional, this helps avoid the rank problem emphasized in the lecture slides: when the return matrix has $T < N$, the sample covariance matrix is singular and cannot be inverted. Thus, 5-minute sampling is preferred to lower-frequency choices such as 15, 30 or 60 minutes, which would leave fewer intraday returns for estimating the 30×30 covariance matrix.

Exercise 2a: Daily realized covariance matrices

We estimate the daily realized covariance matrix for the 30 DJIA stocks using the standard realized covariance estimator

$$RC_d = \sum_{j=1}^{M_d} r_{d,j} r'_{d,j},$$

where $r_{d,j}$ denotes the 30×1 vector of synchronized intraday log returns at interval j on trading day d . This estimator is the multivariate analogue of the realized variance estimator used in Exercise 1.

Following Exercise 1, covariance is estimated using synchronized 5-minute returns. The same sampling frequency is used throughout the remainder of the analysis, ensuring consistency with the realized variance estimator selected based on the volatility signature plot.

To synchronize observations across stocks, prices are placed on a common one-minute calendar grid within each trading day and missing observations are handled using previous-tick interpolation (forward filling). Returns are then computed on a common 5-minute grid, ensuring that the covariance matrix is based on synchronized return vectors.

For each trading day, the realized covariance matrix is obtained as the cross-product of synchronized intraday return vectors.

Exercise 2b: Diagonal elements of the realized covariance matrix

The diagonal elements of the realized covariance matrix RC_d are the daily realized variances of the individual stocks. For stock i , the diagonal element is

$$RC_{d,ii} = \sum_{j=1}^{M_d} r_{d,j,i}^2.$$

Thus, by extracting the diagonal of each daily realized covariance matrix, we obtain a time series of daily realized variance for each of the 30 stocks.

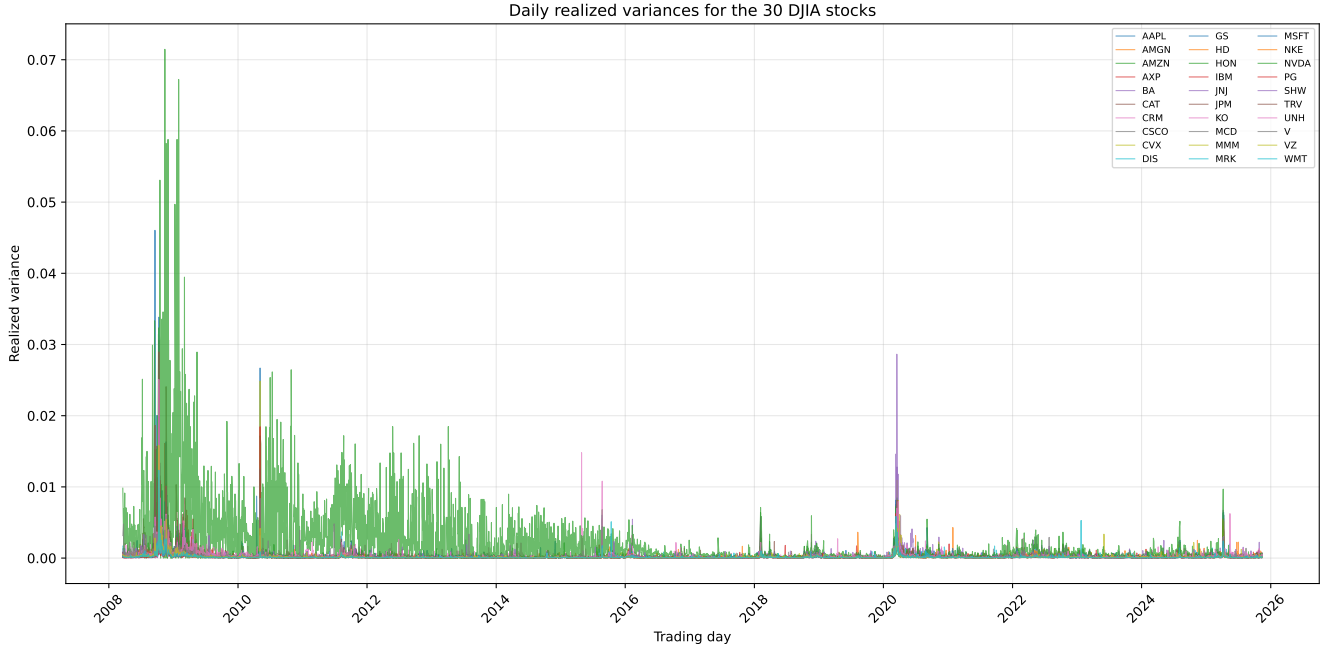


Figure 2: Daily realized variances for the 30 DJIA stocks.

The realized variance series exhibit clear volatility clustering, with periods of high volatility followed by further periods of high volatility and vice versa. Several stocks experience large volatility spikes during periods of market stress, indicating that volatility tends to increase simultaneously across assets.

The figure also shows substantial time variation in realized variance, suggesting that volatility is far from constant over time. While the overall patterns are similar across stocks, some stocks display persistently higher realized variance than others, reflecting differences in underlying risk characteristics.

By sampling at a one-minute frequency instead, the off-diagonal elements of RC_d would likely be more affected by market microstructure noise. As discussed in the lecture slides, Andersen and Bollerslev (1998), and Barndorff-Nielsen et al. (2011), covariance estimates at very high frequencies can suffer from the Epps effect, whereby the measured dependence between assets is biased downward. Although the DJIA data are recorded on a common one-minute grid and therefore largely mitigate the non-synchronous trading problem, one-minute returns remain more sensitive to bid-ask bounce and price discreteness. Consequently, the off-diagonal elements of RC_d may become noisier and potentially lower than those obtained from synchronized five-minute returns. This provides additional support for the five-minute sampling frequency selected in Exercise 1.

Exercise 3: Covariance estimators

For each day t in the second half of the sample, we construct two estimators of today's covariance matrix using only information available up to day $t - 1$.

Exercise 3a: High-frequency random walk estimator

The high-frequency estimator is defined as

$$\hat{\Sigma}_t^{HF} = RC_{t-1}.$$

Thus, yesterday's realized covariance matrix is used as today's covariance forecast. This implies that the covariance matrix follows a random-walk process, such that changes in covariance are unpredictable and arise only from new shocks or information. Consequently, yesterday's realized covariance matrix provides the best forecast of today's covariance matrix. This assumption is motivated by the persistence and clustering typically observed in volatility and covariance dynamics.

Exercise 3b: Ledoit-Wolf shrinkage estimator

For the Ledoit-Wolf estimator, we first construct daily close-to-close log returns for all 30 stocks using the last observed intraday price on each trading day. For each evaluation day t , the Ledoit-Wolf covariance matrix is estimated from a rolling window of $W = 252$ daily returns ending on day $t - 1$. Thus, only information available before day t is used. The resulting matrix is stored as $\hat{\Sigma}_t^{LW}$, and its diagonal elements are saved for the comparison in Exercise 3c.

Exercise 3c: Comparing realized and Ledoit-Wolf variances

To compare the smoothing properties of the two estimators, we focus on the diagonal elements of the covariance matrices. The diagonal elements of $\hat{\Sigma}_t^{LW}$ are the Ledoit-Wolf implied variance forecasts, while the diagonal elements of RC_t are the realized variances computed from intraday returns in Exercise 2b.

Thus, for each stock i , we compare

$$\hat{\Sigma}_{t,ii}^{LW} \quad \text{and} \quad RC_{t,ii}.$$

The Ledoit-Wolf estimator is based on a rolling window of daily close-to-close returns and shrinkage, while RC_t is based on high-frequency intraday returns. The comparison therefore illustrates how much smoothing is introduced by the Ledoit-Wolf estimator relative to the more volatile realized variance series.

Ledoit-Wolf implied variances vs. realized variances

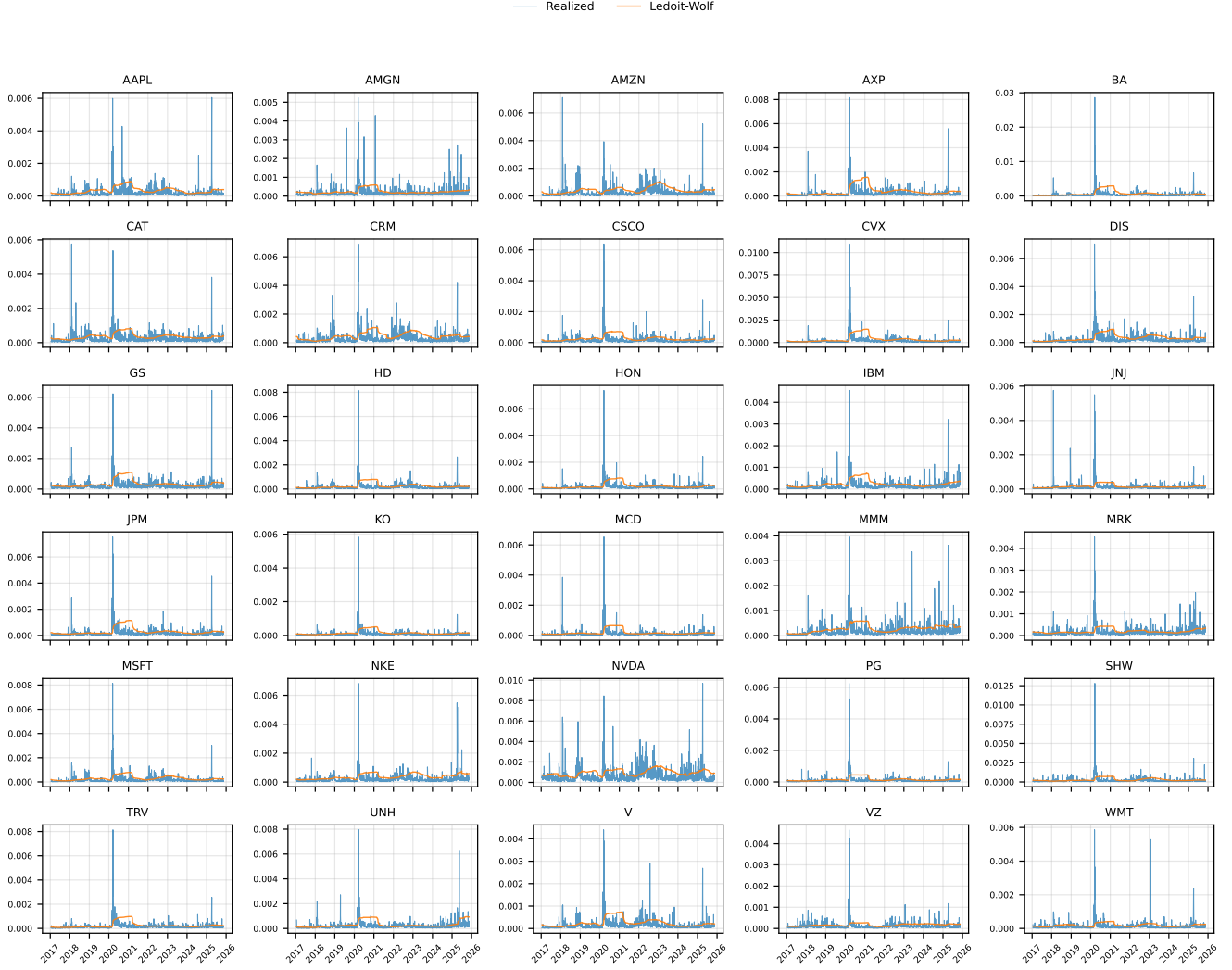


Figure 3: Ledoit-Wolf implied variances and realized variances for the 30 DJIA stocks. The Ledoit-Wolf estimates are based on a 252-day rolling window of daily close-to-close returns, while realized variances are the diagonal elements of the daily realized covariance matrices.

The figure shows that the realized variances are substantially more volatile than the Ledoit-Wolf implied variances. The realized variance series contain sharp daily spikes, especially during periods of market stress, because they are computed directly from intraday returns and therefore react immediately to short-run volatility shocks. In contrast, the Ledoit-Wolf estimates are much smoother and move more gradually over time. This reflects that the estimator is based on a 252-day rolling window of daily close-to-close returns and applies shrinkage to reduce estimation error. As argued by Ledoit and Wolf (2003, 2004), shrinkage pulls noisy and extreme covariance estimates towards a more stable target. The comparison therefore highlights the different smoothing properties of the two estimators: the realized variance estimator exhibits little smoothing and responds immediately to new information, whereas the Ledoit-Wolf estimator provides a smoother and more stable estimate of risk, but reacts more slowly to sudden changes in volatility.

Exercise 4: Minimum-variance portfolios

Using the covariance matrix forecasts constructed in Exercise 3, we form daily minimum-variance portfolios for the evaluation period. Specifically, we compute portfolio weights based on both the high-frequency random-walk estimator, $\hat{\Sigma}_t^{HF}$, and the Ledoit-Wolf shrinkage estimator, $\hat{\Sigma}_t^{LW}$.

For a given covariance matrix estimate $\hat{\Sigma}_t$, the minimum-variance portfolio is obtained by solving

$$\omega_{mv,t} = \arg \min_{\omega_t} \omega_t' \hat{\Sigma}_t \omega_t \quad \text{s.t.} \quad \iota' \omega_t = 1.$$

Solving the first-order conditions yields the closed-form solution

$$\omega_{mv,t} = \frac{\hat{\Sigma}_t^{-1} \iota}{\iota' \hat{\Sigma}_t^{-1} \iota},$$

where ι denotes a vector of ones.

The resulting portfolio weights reflect how the different covariance estimators translate into portfolio allocations. Since the high-frequency estimator is based on the most recent realized covariance matrix, it is expected to react more strongly to short-run fluctuations in covariance. In contrast, the Ledoit-Wolf estimator uses a rolling window of daily returns and shrinkage, which may lead to more stable covariance estimates and therefore more stable portfolio weights. We compare the resulting portfolio allocations, identify the most extreme positions, and evaluate the average turnover between rebalancing dates.

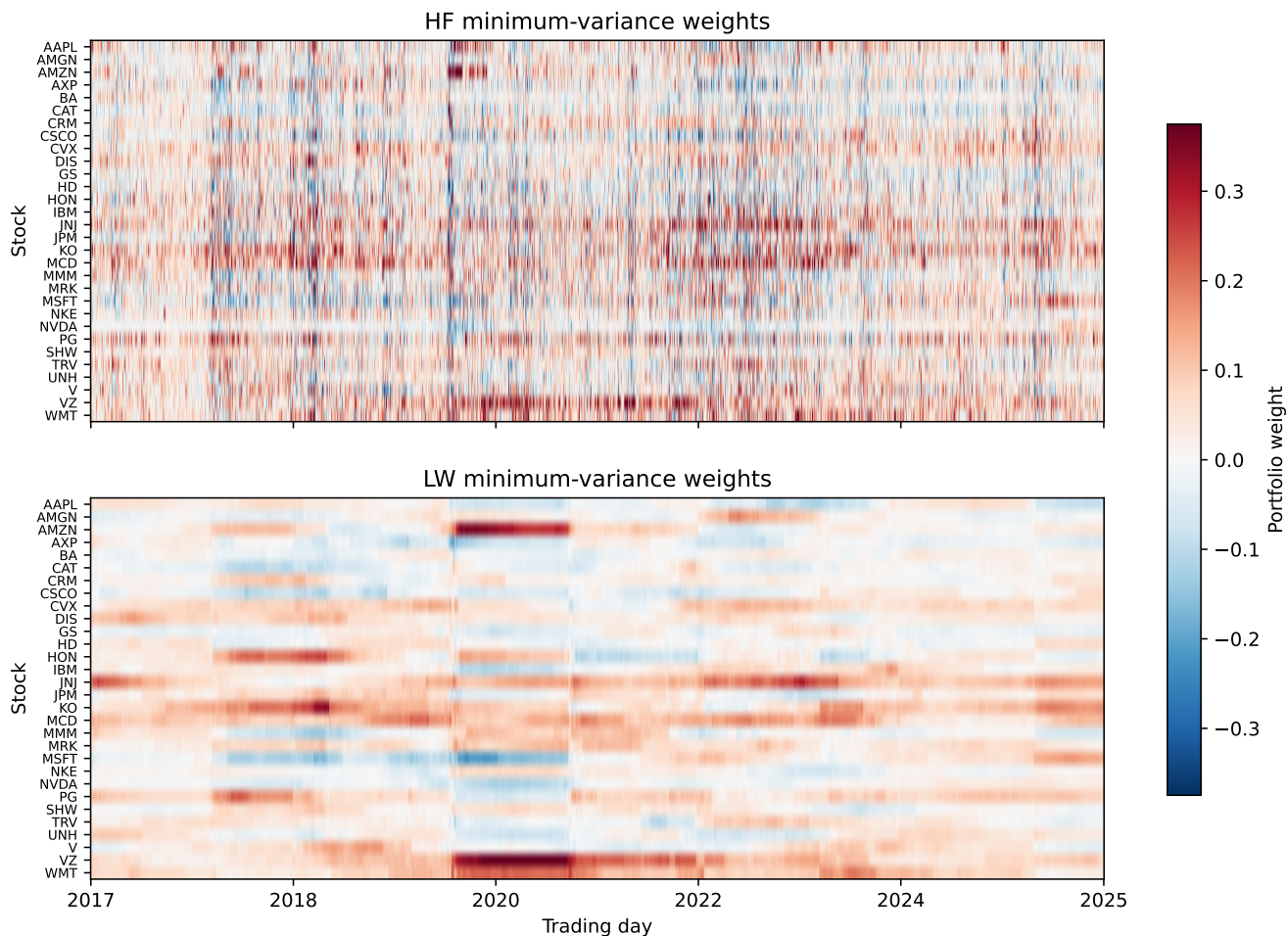


Figure 4: Minimum-variance portfolio weights for the high-frequency and Ledoit-Wolf covariance estimators. Rows are stocks, columns are rebalancing dates. Red indicates long positions, blue indicates short positions, and white indicates weights close to zero. Both panels share the same colour scale, capped at the 99th percentile of absolute weights so that typical variation is visible.

Table 1: Largest absolute minimum-variance portfolio positions.

Trading day	Strategy	Stock	Weight (%)	Absolute weight
2021-10-05 00:00:00	HF	VZ	148.1	1.481
2020-03-30 00:00:00	HF	AMZN	133.97	1.3397
2020-03-09 00:00:00	HF	AMZN	119.76	1.1976
2020-03-19 00:00:00	HF	AAPL	119.02	1.1902
2021-10-01 00:00:00	HF	VZ	118.94	1.1894
2020-03-04 00:00:00	HF	HD	-116.91	1.1691
2020-03-27 00:00:00	HF	AMZN	110.87	1.1087
2020-04-13 00:00:00	HF	AMZN	110.12	1.1012
2023-01-05 00:00:00	HF	JNJ	100.63	1.0063
2021-10-07 00:00:00	HF	VZ	100.23	1.0023

Table 2: Turnover of the high-frequency and Ledoit-Wolf minimum-variance portfolios.

Strat- egy	aver- age_turnover	me- dian_turnover	max_turnover	Rebal- ances	Average turnover (%)	Median turnover (%)	Max turnover (%)
HF	3.49653	3.28441	9.34559	2222	349.65	328.44	934.56
LW	0.0712183	0.0498634	0.984544	2222	7.12	4.99	98.45

Exercise 5

Table 3: Out-of-sample performance of the high-frequency, Ledoit-Wolf, and equal-weighted strategies.

Strategy	Ann. return (%)	Ann. volatility (%)	Ann. Sharpe ratio
EW	17.75	17.87	0.993
HF	11.82	16.58	0.713
LW	11.37	13.99	0.812

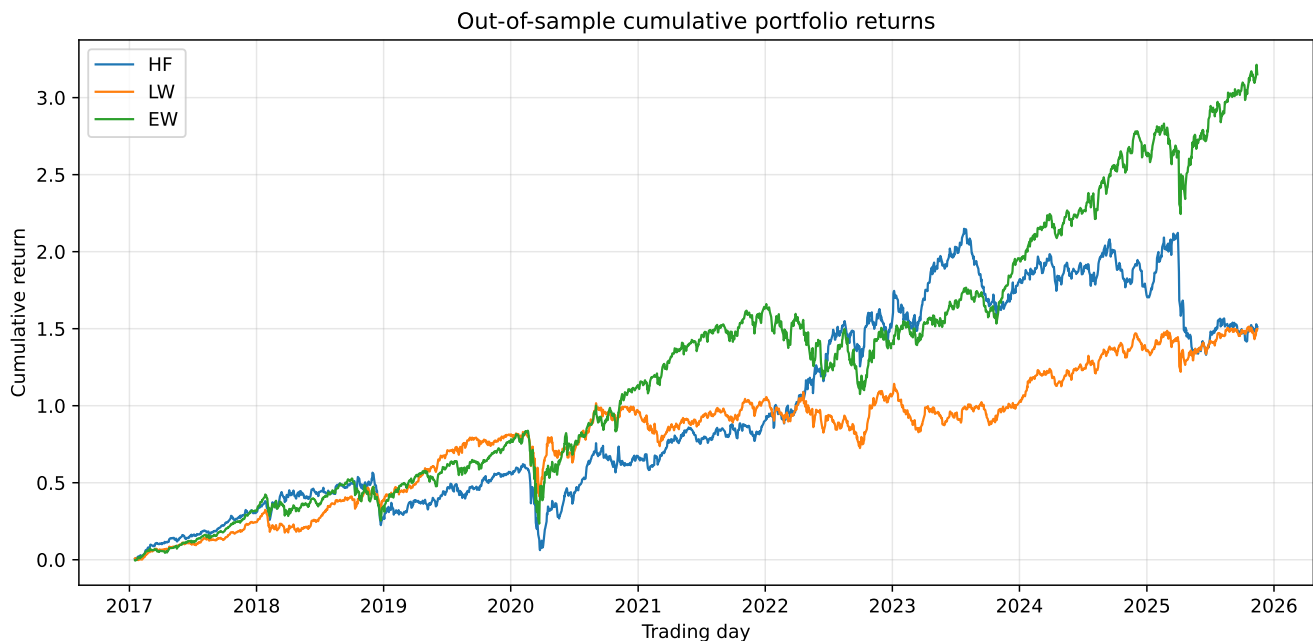


Figure 5: Cumulative out-of-sample portfolio returns for the high-frequency, Ledoit-Wolf, and equal-weighted strategies.

Table 3 reports the out-of-sample results. The Ledoit-Wolf strategy attains the lowest annualized volatility at 13.99% with a Sharpe ratio of 0.812, whereas the high-frequency strategy is both more volatile, at 16.58%, and lower-Sharpe, at 0.713. The equal-weighted benchmark has the highest volatility, 17.87%, but also the highest Sharpe ratio at 0.993.

The high-frequency estimator therefore does not deliver a measurable advantage over Ledoit-Wolf shrinkage (Ledoit and Wolf, 2003, 2004). On the contrary, it is dominated on both dimensions: it is more volatile and has a lower Sharpe ratio than the shrinkage estimator, even before trading costs are taken into account. This is notable given that both are minimum-variance portfolios: Ledoit-Wolf achieves the lowest realized volatility of all three strategies, exactly as the minimum-variance objective intends, while the high-frequency estimator does not. The difference is the same instability documented in Exercise 4, where the high-frequency strategy rebalanced with an average turnover of roughly 349% against only 7% for Ledoit-Wolf. Once realistic transaction costs are accounted for, the small amount of risk the high-frequency estimator does remove would be far outweighed by the cost of its trading, so the richer intraday information it exploits is not translated into an economically meaningful improvement. As seen in Figure 5, the high-frequency strategy also displays the most violent swings in cumulative return, including a sharp drawdown in 2025, while Ledoit-Wolf is the smoothest of the three.

The broader result is that neither optimized strategy beats the naive equal-weighted benchmark on a risk-adjusted basis: the benchmark, which estimates nothing and therefore carries no estimation error, attains the highest Sharpe ratio of the three. Whether an estimator that exploits additional information — intraday returns, shrinkage targets, or factor structures — improves on equal weighting depends on whether its reduction in estimation error outweighs the noise it introduces through the inversion of the covariance matrix. As emphasised in the in-class discussion and in the shrinkage literature (?), the sample covariance matrix is poorly conditioned when the number of assets is large relative to the sample, and imposing structure can substantially improve out-of-sample behaviour; this is precisely why Ledoit-Wolf outperforms the high-frequency estimator here. We would expect sophistication to pay off when the sample is long relative to the number of assets, when the covariance structure is stable and persistent so that past estimates forecast future risk well, and when turnover is controlled or trading costs are low. The minimum-variance problem with thirty assets and a covariance matrix re-estimated daily sits at the opposite end of this spectrum: estimation error dominates, structure (Ledoit-Wolf) helps more than extra information (high-frequency data), and even then the simplest possible rule is hard to beat. Our evidence suggests that statistical sophistication is valuable in portfolio choice mainly insofar as it stabilises the inputs to optimisation, and that sophistication which merely increases responsiveness — as with the daily-updated high-frequency estimator — does not reliably beat a simple benchmark once realistic frictions are considered.

References

- Ledoit, O. and Wolf, M. (2003). Improved estimation of the covariance matrix of stock returns with an application to portfolio selection. *Journal of Empirical Finance*, 10(5):603–621.
- Ledoit, O. and Wolf, M. (2004). Honey, i shrunk the sample covariance matrix. *The Journal of Portfolio Management*, 30(4):110–119.